

Ayush Raj

Third Year Undergraduate
Department of Aerospace Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology Kanpur	6.6/10
2023	CBSE(XII)	Indian Convent School,Kota	94.2%
2021	CBSE(X)	Army Public School,Sri Ganganagar	96.4%

Scholastic Achievements

- Secured All India Rank **3946** in **JEE Advanced 2023** among 2.5 lakh shortlisted candidates.
- Secured All India Rank **4175** in **JEE Mains 2023** among 11 lakh candidates.

Research Experience

SURGE’25 IIT KANPUR | B2B Brand Equity | Prof.RN Sengupta (May’25-July’25)

Objective	Identify determinants of B2B brand equity using literature, company data, and test models via ML .
Approach	- Reviewed literature on B2B Brand Equity to map constructs like awareness, loyalty, trust, and reputation. - Applied SEM and multiple logistic regression on 5 firms; analyzed impact of digital marketing using MROI. - Analysed Semantic Brand Score to measure brand importance and strength using textual data and Monte Carlo Simulation for Brand Equity Risk Optimization
Impact	Proposed methods to quantify brand equity using ML models and highlighted digital marketing’s role via MROI .

Key Projects

White Paper For Assistive Devices | IITK Consulting Group | Department Of SnT (April’25-June’25)

- Worked under **Dr. Anita Aggarwal (DST, NAMEDD)** on a national white paper for assistive devices (ADs), targeting policy investment.
- Co-authored recommendations using global best practices comparative insights from **USA, Japan, Germany, Denmark**.
- Built global leaderboard, mapped **India’s top 10 medtech states**; analyzed **7+ govt schemes, 20+ VC firms/accelera-**

Course Project-DMS624: Computer-Aided Decision Systems | Prof. Deepu Philip (Aug’25-Nov’25)

- OrbitWear: End-to-End Merchandise Decision System**
 - Designed **end-to-end merchandise solution** (design, pricing, delivery) for IITK clubs, resolving vendor coordination issues.
 - Managed **pricing, advance payments, vendor payouts, and profit tracking** under a low-capital model to minimize financial risk.
- Solving Farmers’ Loan Access: Digital Kisan ID–Based Credit Framework**
 - Analyzed credit access barriers using **NABARD, RBI, & PM-KISAN reports** to identify systemic bottlenecks.
 - Proposed **Digital Kisan ID** integrating Aadhaar & land records to **auto-fill 80% of loan forms** for instant application.

TaxSmart | Capstone Project | Product Club IITK (Dec’25)

- Conceptualized a **proactive tax planning product** addressing fragmented data, low tax awareness, and last-minute filing behavior.
- Designed an **MVP** including a Tax Planner Quiz, personalized tax plan, document vault, year-round tracking, and a unified Tax Health Score.
- Conducted competitive analysis of **Groww, ClearTax, and ET Money**; identified personalization gaps and proposed a product-led {GTM with content-led onboarding, reminder-based retention, and **ELSS/NPS/insurance** cross-sell.

Technical Skills

- Programming Languages:** C, C++, Python
- Software:** Canva, MS Excel, Jupyter
- Libraries:** Numpy, Pandas, Tensorflow, Seaborn, Matplotlib, Keras, Scikit-Learn

Positions of Responsibility

Business Head | IITK Rocketry & Space Exploration Team Mar’24-Present

- Contributed to **avionics, structures, propulsion, recovery**; launched rocket to **600 ft** apogee in **July 2024**.
- Managed finances; secured **2 sponsorships**, handled **5 lakh** budget, and oversaw bill clearance.
- Led **30+ juniors**, handled outreach, sponsorship, and participating as Recovery member in **INSPACE Competition**.

Marketing Secretary | Udghosh’24 Aug’24-Oct’24

- Led lead generation and **database management** for **100+ companies** to support sponsorship and marketing outreach.
- Developed strong **communication skills** including cold calling, email writing, and persuasive brand negotiation.

Extra-Curricular Activities

- Completed **NCC ‘C’ Certificate** program, showcasing leadership, discipline, commitment, and physical endurance.
- Represented Team RaSET at **IITK Samanvay: Industry Engagement Event 2025**, showcasing rocketry work to industry and academia.
- Contributed to **Antarikx’25 – National Space Day Celebration**, conducting live demos and science outreach on space.
- Participated in and cleared **PDR and CDR** rounds of **INSPACE Model Rocketry Competition** with IITK RaSET team.